

LR-200 Floor Robot — Technical Manual (revision C)

1. Overview

The LR-200 is a commercial floor-cleaning robot for retail spaces up to 1,200 square metres per charge. It combines a rotating brush, a 4-litre clean-water tank and a 3.5-litre recovery tank. The LR-200 is a fictional product described for demonstration purposes.

Specification	Value
Dimensions (W x D x H)	520 x 610 x 480 mm
Weight (empty tanks)	38 kg
Battery	LiFePO4, 25.6 V, 40 Ah
Runtime per charge	up to 3.5 hours
Full charge time	2 hours 15 minutes
Noise level	58 dB(A)
Max slope	6 degrees

2. Charging

Return the robot to its dock when the battery indicator falls below 15 percent. The robot returns to the dock automatically at 10 percent. Do not charge the robot at ambient temperatures below 5 °C or above 40 °C.

3. Error codes

Code	Meaning	Action
E01	Brush jammed	Power off, remove debris from the brush housing.
E02	Recovery tank full	Empty and rinse the recovery tank.
E04	Cliff sensor blocked	Clean the four cliff sensors with a dry cloth.
E07	Battery over temperature	Move the robot to a cooler area and wait 30 minutes.
E12	Navigation lost	Push the robot back to a mapped area and restart the task.

If an error code repeats more than three times in one shift, log a support ticket with the robot's serial number and the time of each error.

4. Maintenance schedule

Daily: empty and rinse the recovery tank, and wipe the squeegee blades. Weekly: clean the cliff sensors and check the brush for wrapped hair or string. Every 500 operating hours: replace the brush and the squeegee blades. Every 2,000 operating hours: have a certified technician replace the drive belts.

5. Warranty

The LR-200 has a 24-month limited warranty covering manufacturing defects. The battery is covered for 36 months or 1,500 charge cycles, whichever comes first. The warranty does not cover damage caused by non-approved cleaning chemicals.